

SEVERITY & CASE CLASSIFICATION

Pharmacovigilance Assignment Report

Task ID	task4	Subject	Pharmacovigilance
Prepared by	Student	Report type	Educational case analysis

Report overview: This report explains how adverse drug reactions are classified according to severity, seriousness, and expectedness. Three illustrative cases are analysed to determine their possible clinical impact.

1. Introduction

Severity and case classification are important parts of pharmacovigilance. They help healthcare professionals understand the clinical impact of an adverse drug reaction (ADR), decide how urgently it should be managed, and determine whether the event was expected from the known safety profile of the medicine.

2. Aim

To classify adverse drug reactions according to severity, seriousness, and expectedness, and to evaluate their clinical impact using basic pharmacovigilance principles.

3. Objectives

- To understand the difference between severity and seriousness.
- To identify mild, moderate, and severe adverse drug reactions.
- To classify reactions as serious or non-serious.
- To understand expected and unexpected adverse reactions.
- To evaluate different patient cases and determine their clinical impact.

4. Important Definitions

Term	Meaning
Severity	The intensity of symptoms or the reaction, such as mild, moderate, or severe.
Seriousness	A regulatory outcome based on events such as death, life-threatening condition, hospitalisation, disability, congenital anomaly, or another medically important condition.
Expected reaction	A reaction consistent with the information already known about the medicine.

Term	Meaning
Unexpected reaction	A reaction whose nature, severity, or outcome is not consistent with available product safety information.

5. Severity Classification

Severity	Description	Example
Mild	Symptoms are minor and usually do not require special treatment or interruption of therapy.	Mild nausea or slight headache.
Moderate	Symptoms are more noticeable and may require treatment, dose adjustment, or temporary interruption.	Persistent vomiting or significant dizziness.
Severe	Symptoms are intense, disabling, or may require urgent medical care.	Severe allergic reaction or severe bleeding.

6. Seriousness Classification

- Death or a life-threatening condition.
- Hospitalisation or prolongation of existing hospitalisation.
- Persistent or significant disability or incapacity.
- Congenital anomaly or birth defect.
- Another medically important event requiring intervention to prevent a serious outcome.

Important distinction: A reaction may be severe in intensity but not meet the regulatory definition of serious. Similarly, an event may be serious because it requires hospitalisation even if the symptoms are not described as severe.

7. Case Studies and Classification

Cas e	Drug and reaction	Classification	Clinical impact
01	Amoxicillin – mild skin rash and itching; no hospitalisation.	Mild, non-serious, known type of reaction.	Low impact; monitor and provide appropriate advice.
02	Ibuprofen – gastrointestinal bleeding requiring hospitalisation.	Severe, serious, potentially expected based on known risk.	High impact; urgent evaluation and treatment required.
03	Metformin – lactic acidosis with serious clinical deterioration.	Severe, serious, potentially life-threatening.	Very high impact; emergency management required.

8. Case 1: Amoxicillin

A patient develops a mild skin rash and itching after starting Amoxicillin. The patient is stable, does not require hospitalisation, and symptoms are manageable.

Assessment: The reaction is mild in severity and non-serious. It is compatible with a known type of reaction to the medicine. The clinical impact is low, but the patient should be assessed and advised appropriately.

9. Case 2: Ibuprofen

A patient taking Ibuprofen develops gastrointestinal bleeding and requires hospitalisation.

Assessment: The reaction is severe and serious because hospitalisation is required. Gastrointestinal bleeding is a clinically important known risk of NSAIDs. The case requires immediate medical evaluation, review or withdrawal of the suspected drug, and appropriate treatment.

10. Case 3: Metformin

A patient taking Metformin develops lactic acidosis with marked clinical deterioration.

Assessment: The event is severe and serious, and potentially life-threatening. It requires emergency medical management. Expectedness should be checked against current product information and the patient's clinical risk factors.

11. General Pharmacovigilance Approach

- Record the patient's age, medical history, and all medicines being used.
- Describe the reaction, onset time, duration, and outcome.
- Assess severity and seriousness separately.
- Check whether the event is expected from approved safety information.
- Identify the suspected medicine and any alternative causes.
- Report serious, unexpected, or clinically important reactions through the appropriate pharmacovigilance system.

12. Conclusion

Severity describes the intensity of an adverse reaction, whereas seriousness describes its clinical and regulatory importance. Expectedness is assessed by comparing the event with established product safety information. In the illustrative cases, Amoxicillin caused a mild non-serious reaction, Ibuprofen was associated with serious gastrointestinal bleeding, and Metformin was associated with a severe potentially life-threatening event.

13. Note

This is a student-prepared educational report based on the Task 4 description. The cases are illustrative and are not real patient records. Classification in an actual report should be performed by a qualified healthcare professional using current pharmacovigilance guidance and product information.

14. References

- World Health Organization (WHO): Pharmacovigilance and adverse drug reaction monitoring.
- Standard textbooks of Pharmacology, Clinical Pharmacy, and Pharmacovigilance.

- Current product information and standard adverse drug reaction reporting principles.